

Data Mining Numerical Problems & Solutions

CT 72502 (Elective I) · BEI / BEX IV-I · TU, IOE · Chapter 4: Association Analysis

Ch 4 – Association Analysis – Numerical Problems & Solutions

5 methods · 14 PYQ instances · every itemset count recomputed from the printed transaction tables. **This is the heaviest numerical chapter in the syllabus.**

Methods

- **M1** Apriori: candidate generation, pruning, frequent itemsets
- **M2** Rule generation from frequent itemsets
- **M3** FP-tree construction and FP-Growth mining
- **M4** Pattern evaluation: lift and the correlation measures
- **M5** Subsequence enumeration

Convert the threshold to a COUNT before you start. “min support 50%” on 6 transactions means a count of 3. Nearly every lost mark in this chapter comes from mixing percentages and counts half way through. Write $\text{min_sup_count} = N$ at the top and use only that.

Also: **support is over ALL transactions; confidence is over the antecedent's transactions only.**

M1. Apriori

Method

1. Scan once, count items $\rightarrow$ keep those $\geq \text{min_sup_count} \rightarrow L_1$.
 2. **Join** L_{k-1} with itself: combine 2 itemsets only if their first $k-2$ items match.
 3. **Prune**: discard a candidate if *any* of its $(k-1)$ -subsets is missing from L_{k-1} .
 4. **Count** survivors against D , keep those meeting the threshold $\rightarrow L_k$.
 5. Stop when L_k is empty.
- **Show the prune step explicitly** – it is where the method marks are.

Problem 1.1 2+7

80 Bh 7 transactions, min support count = 4, confidence 80%. Note T106 prints “F, A, C, I, B, A, P” — **A is repeated; a transaction is a set**, so treat it as {F,A,C,I,B,P}.

Step 1: item counts

(7 transactions)

- A:6 · P:6 · B:5 · C:5 · F:5 · M:4 · I:3 · V:3 · D:2 · L:2 · N:2 · O:2 · S:2 · others 1
- $L_1 = \{\mathbf{A(6)}, \mathbf{B(5)}, \mathbf{C(5)}, \mathbf{F(5)}, \mathbf{M(4)}, \mathbf{P(6)}\}$

Step 2: L_2

(15 candidate pairs; counts shown, ≥ 4 kept)

- Kept: **AB(4), AC(5), AF(4), AM(4), AP(5), BP(4), CF(4), CP(4), FP(5), MP(4)**
- Dropped: BC(3), BF(3), BM(2), CM(3), FM(3)

Step 3: L_3

- ABP: all 3 subsets present $\rightarrow$ counted = 3 ☐
- **ACF(4), ACP(4), AFP(4), AMP(4), CFP(4)** ☐
- Pruned without counting: ABC ($BC \notin L_2$), ABF (BF), ABM (BM), ACM (CM), AFM (FM)

Step 4: L_4

- ACFP: subsets ACF ☐ ACP ☐ AFP ☐ CFP ☐ $\rightarrow$ candidate. Count = 4 (T100, T101, T104, T106) ☐
- $L_4 = \{\mathbf{ACFP(4)}\}$. L_5 empty $\rightarrow$ stop.

Final answer

Maximal frequent itemset = **{A, C, F, P}** with support $4/7 = 57\%$.

- **Check:** all $2^4 - 2 = 14$ proper subsets of ACFP do appear in $L_1 - L_3$ above. ✓

Problem 1.2 5

80 Ba 4 transactions, min support 2.

- L_1 : A(2), B(3), C(3), E(3) (D has count 1, dropped)
- $C_2 \rightarrow L_2$: **AC(2), BC(2), BE(3), CE(2)**; dropped AB(1), AE(1)
- C_3 : BCE (subsets BC, BE, CE all present) $\rightarrow$ count = 2 ✓. AC joins with nothing.
- $L_3 = \{\mathbf{BCE(2)}\}$

Final answer

Frequent: A, B, C, E, AC, BC, BE, CE, BCE.

Problem 1.3 10

70 Ch 4 transactions (ACD, BD, ABCE, BDF), support 50% $\rightarrow$ count 2.

- L_1 : A(2), B(3), C(2), D(3) (E:1, F:1 dropped)
- C_2 : AB(1) ✓ AC(2) ✓ AD(1) ✓ BC(1) ✓ BD(2) ✓ CD(1) ✓
- $L_2 = \{\mathbf{AC(2), BD(2)}\}$
- C_3 : AC and BD share no common prefix $\rightarrow$ **no candidates**. Stop.

Final answer

Frequent itemsets: A, B, C, D, AC, BD.

Problems 1.4 and 1.5 5+3 8

76 Ch (min sup 2/9, min conf 7/9) and 74 Ash (min sup 20%, min conf 80%). **Identical 9-transaction table, and both thresholds work out to a count of 2**, so the frequent itemsets are the same; only the rule filter differs.

Frequent itemsets

(9 transactions, min count 2)

- L_1 : M1(6), M2(7), M3(6), M4(2), M5(2)
- L_2 : **M1M2(4), M1M3(4), M1M5(2), M2M3(4), M2M4(2), M2M5(2)**; dropped M1M4(1), M3M4(0), M3M5(1), M4M5(0)
- L_3 : **M1M2M3(2), M1M2M5(2)**. Pruned: M1M2M4 (M1M4 ✓), M2M3M5 (M3M5 ✓), M1M3M5 (M3M5 ✓)
- L_4 : M1M2M3M5 pruned (M1M3M5 ✓) $\rightarrow$ empty.

1.4 – rules $X \wedge Y \Rightarrow Z$, min conf 7/9 = 0.778

- From M1M2M3 (sup 2): all three give $2/4 = 0.5$ ✓
- From M1M2M5 (sup 2):
 - $M1 \wedge M2 \Rightarrow M5 = 2/4 = 0.5$ ✓
 - $M1 \wedge M5 \Rightarrow M2 = 2/2 = 1.0$ ✓
 - $M2 \wedge M5 \Rightarrow M1 = 2/2 = 1.0$ ✓

Answer

2 strong rules, both at 100% confidence.

1.5 – all rules, min conf 0.80

- $M5 \Rightarrow M1 = 2/2 = 1.0$ ✓ $M5 \Rightarrow M2 = 2/2 = 1.0$ ✓
- $M4 \Rightarrow M2 = 2/2 = 1.0$ ✓
- $M5 \Rightarrow M1 \wedge M2 = 2/2 = 1.0$ ✓
- Plus the two 3-itemset rules from 1.4.
- All rules with M1, M2 or M3 as antecedent fail (best is $4/6 = 0.667$).
- **Pattern:** only the *rare* items (M4, M5) produce confident rules – they never appear without their partners.

Problem 1.6 8

79 Bh 6 transactions, min support 50%, min confidence 80%. **This one has a surprise.**

- 6 transactions, 50% $\rightarrow$ min count = 3.
- L_1 : A(4), B(3), E(3) (C:2, D:2)
- C_2 : AB = 1, AE = 1, BE = 2
- $L_2 = \emptyset$

Final answer

- **No frequent 2-itemset exists, so no association rule can be derived at 50% support.** Say this explicitly – it is the correct answer, not a dead end.
- **Show you understand why:** the data is spread thin, every pair occurs at most twice.
- **If min support is relaxed to 33.3% (count 2):** $L_2 = \{AC(2), BE(2)\}$, and the only rule clearing 80% confidence is $C \Rightarrow A$ at $2/2 = 100\%$. Worth adding as a one-line remark. verified/added

M2. Rule generation from frequent itemsets**Method**

- For each frequent itemset l with $|l| \geq 2$, take every non-empty proper subset s :

$$s \Rightarrow (l - s) \quad \text{with} \quad \text{conf} = \frac{\text{sup_count}(l)}{\text{sup_count}(s)}$$

- Keep it if $\text{conf} \geq \text{min_conf}$. A k -itemset gives $2^k - 2$ candidate rules.
- **The support of every rule from l is the same** ($= \text{sup}(l)$); only confidence varies.
- **Shortcut:** if $s \Rightarrow (l - s)$ fails, every rule whose antecedent is a subset of s also fails. Prune downward.

Worked example

	Rule	Confidence
From $l = \{K, E, O\}$ with $\text{sup} = 3$ (see Problem 3.1), $ l = 3 \rightarrow 6$ candidate rules:	$K \Rightarrow E, O$	3/5
	$E \Rightarrow K, O$	3/4
	$O \Rightarrow K, E$	3/3
	$K, E \Rightarrow O$	3/4
	$K, O \Rightarrow E$	3/3
	$E, O \Rightarrow K$	3/3

- Every rule with **O** in the antecedent succeeds, because O never appears without K and E.

M3. FP-tree construction and FP-Growth mining**Method**

- Scan 1:** count items, drop infrequent ones, sort the rest by descending support $\rightarrow$ **F-list**. **State your tie-break rule.**
- Scan 2:** rewrite each transaction in F-list order, insert as a path, sharing prefixes and incrementing counts. Build the header table with node-links.
- Mine bottom-up** (least frequent item first): conditional pattern base $\rightarrow$ conditional FP-tree $\rightarrow$ recurse.

Problem 3.1 2+7

81 Ba 5 transactions, min support 60%, min confidence 80%. Find all frequent itemsets using FP-growth, then all strong rules.

Given

T100 {M, O, N, K, E, Y} T200 {D, O, N, K, E, Y} T300 {M, A, K, E} T400 {M, U, C, K, Y} T500 {C, O, O, K, I, E}

- **T500 prints O twice** – a transaction is a set, so it is {C, O, K, I, E}.
- 5 transactions, 60% $\rightarrow$ **min count = 3**.

Step 1: counts and F-list

- K:5, E:4, M:3, O:3, Y:3 N:2, C:2, D:1, A:1, U:1, I:1
- **F-list = K(5), E(4), M(3), O(3), Y(3)** (ties at 3 broken alphabetically)

Step 2: reordered transactions

- T100 → K, E, M, O, Y
- T200 → K, E, O, Y
- T300 → K, E, M
- T400 → K, M, Y
- T500 → K, E, O

Step 3: the FP-tree

- root
- → K:5
- → E:4
- → M:2 → O:1 → Y:1
- → O:2 → Y:1
- → M:1 → Y:1
- **Check:** counts on each level sum correctly – K:5 = E:4 + M:1. ✓

All frequent itemsets

- 1-itemsets: K:5, E:4, M:3, O:3, Y:3
- 2-itemsets: {K,E}:4, {K,M}:3, {K,O}:3, {E,O}:3, {K,Y}:3
- 3-itemsets: {K,E,O}:3

Strong rules (min conf 80%)

- $K \Rightarrow E = 4/5 = 0.80 \geq 0.8$ $E \Rightarrow K = 4/4 = 1.0 \geq 0.8$
- $M \Rightarrow K = 3/3 = 1.0 \geq 0.8$ $O \Rightarrow K = 3/3 = 1.0 \geq 0.8$ $Y \Rightarrow K = 3/3 = 1.0 \geq 0.8$ $O \Rightarrow E = 3/3 = 1.0 \geq 0.8$
- $K, O \Rightarrow E = 1.0 \geq 0.8$ $E, O \Rightarrow K = 1.0 \geq 0.8$ $O \Rightarrow K, E = 1.0 \geq 0.8$
- Failing: $K \Rightarrow M$, $K \Rightarrow O$, $K \Rightarrow Y$ (all $3/5 = 0.6$), $E \Rightarrow O$ (0.75), $K, E \Rightarrow O$ (0.75), $E \Rightarrow K, O$ (0.75).

Problem 3.2 4+4

75 Ch Draw the FP-tree for the 5-transaction table. Min support is not printed; use 3, which is the standard for this dataset. verified/added

Counts and F-list

- f:4, c:4, a:3, b:3, m:3, p:3 l:2, o:2, and d,g,i,h,j,k,s,e,n,w all 1
- F-list = f, c, a, b, m, p
- **T05 prints a capital "A"** – same item as a.

Reordered transactions

- 01 → f, c, a, m, p 02 → f, c, a, b, m
- 03 → f, b 04 → c, b, p
- 05 → f, c, a, m, p

Step 4: mine bottom-up

Item	Conditional pattern base	Frequent patterns
Y	{K,E,M:1}, {K,E:1}, {K,M:1} counts K:3, E:2, M:2	{K,Y}:3
O	{K,E,M:1}, {K,E:2} counts K:3, E:3	{K,O}:3, {E,O}:3, {K,E,O}:3
M	{K,E:2}, {K:1} counts K:3, E:2	{K,M}:3
E	{K:4}	{K,E}:4
K	–	{K}:5

The FP-tree

- root
- → f:4
- → c:3 → a:3
- → m:2 → p:2
- → b:1 → m:1
- → b:1
- → c:1 → b:1 → p:1
- **Note the second root child:** transaction 04 starts with c, not f, so it cannot share the main prefix.

Problem 3.3 4+4

78 Bh Grocery store, support 33.34%, confidence 60%. Show how the tree evolves, then mine it.

- 6 transactions, 33.34% → **min count = 2**.
- Counts: HotDogs 4, Chips 4, Coke 3, Buns 2, Ketchup 2 — **all frequent**.
- **F-list = Chips(4), HotDogs(4), Coke(3), Buns(2), Ketchup(2)** (4-way tie broken alphabetically).

Tree, built transaction by transaction

- T1 (HotDogs, Buns, Ketchup): $\text{root} \rightarrow \text{H}:1 \rightarrow \text{Bu}:1 \rightarrow \text{Ke}:1$
- T2 (HotDogs, Buns): $\text{H}:2, \text{Bu}:2$
- T3 (Chips, HotDogs, Coke): new branch
 $\text{Ch}:1 \rightarrow \text{H}:1 \rightarrow \text{Co}:1$
- T4 (Chips, Coke): $\text{Ch}:2$, new child $\text{Co}:1$
- T5 (Chips, Ketchup): $\text{Ch}:3$, new child $\text{Ke}:1$
- T6 (Chips, HotDogs, Coke): $\text{Ch}:4, \text{H}:2, \text{Co}:2$

Final tree

- root
- → Chips:4 → {HotDogs:2 → Coke:2, Coke:1, Ketchup:1}
- → HotDogs:2 → Buns:2 → Ketchup:1

Mining

Item	Cond. pattern base	Patterns
Ketchup	{H,Bu:1}, {Ch:1}	none (all counts 1)
Buns	{H:2}	{H,Bu}:2
Coke	{Ch,H:2}, {Ch:1}	{Ch,Co}:3, {H,Co}:2, {Ch,H,Co}:2
HotDogs	{Ch:2}	{Ch,H}:2
Chips	—	{Ch}:4

Frequent itemsets

- Chips 4, HotDogs 4, Coke 3, Buns 2, Ketchup 2
- {Chips,Coke} 3, {Chips,HotDogs} 2, {HotDogs,Coke} 2, {HotDogs,Buns} 2
- {Chips,HotDogs,Coke} 2

Problem 3.4 2+8

74 Ch Electronics store, 6 transactions. Min support not printed; use 2.

- Counts: Laptop 4, Earphone 4, Mobile 3, Camera 2, Pen drive 2 — **all frequent**.
- **F-list = Earphone(4), Laptop(4), Mobile(3), Camera(2), Pen drive(2)**
- Reordered: T1 (L, Ca, Pd) · T2 (L, Pd) · T3 (E, L, Mo) · T4 (E, Mo) · T5 (E, Ca) · T6 (E, L, Mo)
- **Tree:** $\text{root} \rightarrow \text{Earphone}:4 \rightarrow \{\text{Laptop}:2 \rightarrow \text{Mobile}:2, \text{Mobile}:1, \text{Camera}:1\}$; and $\text{root} \rightarrow \text{Laptop}:2 \rightarrow \{\text{Camera}:1 \rightarrow \text{PenDrive}:1, \text{PenDrive}:1\}$
- Key frequent pattern: **{Earphone, Laptop, Mobile}:2**

M4. Pattern evaluation: lift and correlation**Method**

$$\text{lift}(A, B) = \frac{P(A \cup B)}{P(A)P(B)} = \frac{\text{conf}(A \Rightarrow B)}{\text{sup}(B)} \quad \chi^2 = \sum \frac{(\text{obs} - \text{exp})^2}{\text{exp}}$$

- $\text{lift} > 1$ positive · = 1 independent · < 1 negative.

Problem 4.1 8

70 Ch “Why is pattern evaluation important? Explain with example the statistical based measures.” Use the standard game/video contingency table.

verified/added

Given

10,000 transactions: 6,000 contain *game*, 7,500 contain *video*, 4,000 contain both.

Step 1: the support–confidence view says the rule is strong

- $\text{sup}(\text{game} \Rightarrow \text{video}) = 4000/10000 = 40\%$ (passes a 30% threshold)
- $\text{conf}(\text{game} \Rightarrow \text{video}) = 4000/6000 = 66.7\%$ (passes a 60% threshold)

Step 2: but look at the consequent alone

- $P(\text{video}) = 7500/10000 = 75\%$
- **Buying a game DROPS the chance of buying a video from 75% to 66.7%.** The rule is actively misleading.

Step 3: lift catches it

$$lift = \frac{0.40}{0.60 \times 0.75} = \frac{0.40}{0.45} = 0.89 < 1 \rightarrow \text{negatively correlated}$$

Step 4: χ^2 agrees

	game	<u>game</u>	row
video	4000 (4500)	3500 (3000)	7500
<u>video</u>	2000 (1500)	500 (1000)	2500
col	6000	4000	10,000

$$\chi^2 = \frac{500^2}{4500} + \frac{500^2}{3000} + \frac{500^2}{1500} + \frac{500^2}{1000} = 55.6 + 83.3 + 166.7 + 250 = 555.6$$

- Large $\chi^2 \rightarrow$ correlated; observed (4000) < expected (4500) $\rightarrow$ **negatively**.

Final answer

- **Pattern evaluation matters because support and confidence alone certify rules that are false.** Add a correlation measure before reporting any rule.
- Mention that lift and χ^2 are **not null-invariant**, so for sparse data prefer **all_conf**, **max_conf**, **Kulczynski** or **cosine**.

M5. Subsequence enumeration

Method

- A **k -subsequence** contains k items in total (not k elements).
- Choose how many items to take from each element, keeping element order; a chosen element must be non-empty to remain in the sequence.
- **Enumerate systematically** by the split (a, b, c, d) of items taken per element, then remove duplicates.

Problem 5.1 3

76 Ch List all 4-subsequences of $s = \langle \{1, 3\} \{2\} \{2, 3\} \{4\} \rangle$.

Given

$e_1 = \{1, 3\}$, $e_2 = \{2\}$, $e_3 = \{2, 3\}$, $e_4 = \{4\}$; 6 items in total.

Step 1: enumerate the item-count splits

	(a, b, c, d)	Ways	Subsequences
	(2,1,1,0)	2	$\langle \{1, 3\} \{2\} \{2\} \rangle$, $\langle \{1, 3\} \{2\} \{3\} \rangle$
	(2,1,0,1)	1	$\langle \{1, 3\} \{2\} \{4\} \rangle$
	(2,0,2,0)	1	$\langle \{1, 3\} \{2, 3\} \rangle$
(a, b, c, d) with $a \leq 2, b \leq 1, c \leq 2, d \leq 1$, sum = 4	(2,0,1,1)	2	$\langle \{1, 3\} \{2\} \{4\} \rangle$ (duplicate), $\langle \{1, 3\} \{3\} \{4\} \rangle$
	(1,1,2,0)	2	$\langle \{1\} \{2\} \{2, 3\} \rangle$, $\langle \{3\} \{2\} \{2, 3\} \rangle$
	(1,1,1,1)	4	$\langle \{1\} \{2\} \{2\} \{4\} \rangle$, $\langle \{1\} \{2\} \{3\} \{4\} \rangle$, $\langle \{3\} \{2\} \{2\} \{4\} \rangle$, $\langle \{3\} \{2\} \{3\} \{4\} \rangle$
	(1,0,2,1)	2	$\langle \{1\} \{2, 3\} \{4\} \rangle$, $\langle \{3\} \{2, 3\} \{4\} \rangle$
	(0,1,2,1)	1	$\langle \{2\} \{2, 3\} \{4\} \rangle$

Final answer

- 15 selections, but $\langle \{1, 3\} \{2\} \{4\} \rangle$ arises twice (the $\{2\}$ can come from e_2 or e_3), so there are **14 distinct 4-subsequences**.
- **The exam usually only wants "at least four"** 79 Bh, so listing the first few from a clearly stated method is enough. **Show the method, not just the list.**

Problem 5.2 3

79 Bh "Generate at least four subsequences from $\langle \{2, 3, 5\}, \{6, 7, 8\}, \{9, 1\}, \{7, 4\} \rangle$."

- Any order-preserving selection works. Four valid examples:
 - $\langle \{2\} \{6\} \rangle$ $\langle \{3, 5\} \{7\} \{9\} \rangle$
 - $\langle \{2, 3\} \{6, 7\} \{1\} \{4\} \rangle$ $\langle \{5\} \{8\} \{9, 1\} \{7\} \rangle$
- **Must hold:** element order preserved, and each chosen sub-element is a subset of the original element. **Must not do:** reorder elements, or merge items from two elements into one.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
DM	data mining	DW	data warehouse
DB	database	DBMS	database management system
KDD	knowledge discovery in databases	DSS	decision support system
OLTP	on-line transaction processing	OLAP	on-line analytical processing
ROLAP	relational OLAP	MOLAP	multidimensional OLAP
HOLAP	hybrid OLAP	ETL	extract, transform, load
BI	business intelligence	ER	entity–relationship
attr(s)	attribute(s)	dim(s)	dimension(s)
multidim	multidimensional	txn(s)	transaction(s)
info	information	ops	operations
repo	repository	freq	frequent / frequency
config	configuration	no.	number
PCA	principal component analysis	SVD	singular value decomposition
SMC	simple matching coefficient	KNN	<i>k</i> -nearest neighbour
ANN	artificial neural network	POS	point of sale
FK	foreign key	UI	user interface
GIGO	garbage in, garbage out	w/, w/o	with, without
rm	remove	vs	versus
Eg	example / for example	+Fig	draw a figure
→	leads to / therefore	+	and

Exam-paper month codes (used in the year tags): Ba = Baishakh · Jth = Jestha · Asa = Ashar · Shr = Shrawan · Bh = Bhadra · Ash = Ashwin · Ka = Kartik · Mng = Mangsir · Po = Poush · Ma = Magh · Ch = Chaitra.

Tier chips: **TOP /17** = topic asked in 7 or more of the 17 papers · **HOT /17** = 4–6 papers · **PIN /17** = 1–3 papers but worth ≥ 8 marks. **Mark chips** such as **2+5** are the marks that question actually carried. **verified/added** marks anything researched and added that was not in the source notes.